

The key benefits and features of KiwiSDR are:

- AM FM SSB modes in radio frequency analysis
- Live radio waterfall analysis
- Signal waterfall capturing and evaluations
- Audio recording option
- CW decoding support
- Cross-platform compatibility
- DRM decoding support
- Shortwave/medium wave signals intelligence
- Educational remote labs
- Wide frequency range
- GPS disciplined timing
- High frequency stability
- IQ data streaming
- Low latency streaming
- Multi-user support
- No local hardware
- Public global receivers
- Real-time waterfall display
- Remote HF monitoring
- Shared research access
- TDoA geolocation feature
- Web browser access

- **Wideband spectrum view**
Cloud-based SDR platforms integrate real-time frequency tuning, spectrum waterfalls, modulation selection and recording functionalities without requiring physical deployment of hardware. Researchers and radio signals enthusiasts can analyse HF, VHF, AM, FM, and many other bands remotely without physical devices or dongles. These platforms are useful for propagation studies, global signal monitoring, radio signals

analysis and collaborative research for different domains. Cloud-based SDR deployments and solutions reduce infrastructure cost, offer the ease of shared access, and provide scalable data storage for long-term signal logging and analytics for varied applications.

Governments and defence services can deploy distributed SDR nodes connected to centralised cloud servers for wide-area spectrum surveillance and detection of illegal use of radio frequencies. Academic institutions can



Figure 3: Accessing real-time radio spectrum using cloud KiwiSDR

Table 3: Local SDR vs cloud-based SDR

Cloud SDR platform	Local SDR setup	Practical outcome
Remote browser access	Physical hardware required	Infrastructure difference
Shared cost model	Higher initial investment	Budget variation
Global receiver access	Single location access	Geographic flexibility
Streamed IQ data	Real-time direct sampling	Latency variation
Server-side upgrades	Hardware upgrades needed	Scalability approach
Limited hardware control	Full hardware control	Experiment scope difference
Fixed remote antennas	Custom antenna options	Signal diversity impact
Cloud-based storage	Local data storage	Data management model
Internet dependent	Low network dependency	Connectivity requirement
Remote troubleshooting	On-site troubleshooting	Support mechanism
Multi-country coverage	Limited coverage range	Propagation studies
Centralised security control	Security self-managed	Risk management
Remote power managed	Power supply dependency	Operational continuity
Provider-managed system	Manual maintenance	Maintenance effort
Shared user access	Private experimentation	Usage environment
Ideal for monitoring	Ideal for RF testing	Application focus
Observation-oriented analysis	Custom protocol development	Research orientation
Minimal hardware knowledge	Hardware skill required	User skill requirement